Dohyun (Eric) Kim

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EDUCATION

University of Auckland — **Auckland, New Zealand**

Bachelor of Engineering (Honours) in Computer Systems Engineering — *Expected Graduation Nov 2027*

- **Co-Founder & Academic Team Executive** of *Korean Engineering Body (KEB)* — *Jul 2024 - Oct 2026*

EXPERIENCE

Research Assistant Trainee — **Auckland, New Zealand**

University of Auckland — *Jul 2026 - Oct 2026*

- Trainee researcher analysing the **robot soccer system** at UoA's CARES Lab, covering camera calibration, **vision processing and decision strategy design**, to support its planned migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms and implemented LiDAR integration and control using **ROS and Python**, building hands-on experience in autonomous systems.

Robotics Instructor — **Auckland, New Zealand**

Creative Imaginary Lab (ciLab) — *Apr 2026 - Oct 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

PROJECTS

BFS PhotonTracker — **PSoC & PCB-SMT**

Autonomous Pathfinding Robot - Team Project — *Oct 2026*

- Designed and built a **six sensor light board** that detects a projected line by its 60 Hz flicker, using **LTSpice, PCB design and a PSoC 5** sampling at 4 kHz, with a 3.1 Hz high pass filter to block ambient light.
- Programmed **BFS shortest path planning** and motor speed control in **embedded C**, using a **MATLAB** model to verify paths on 46 routes, so the robot could **navigate a projected maze.**

Flappy Universe — **VHDL & FPGA**

FPGA Game Implementation - Team Project — *Jun 2026*

- Implemented a Flappy Bird style game in **VHDL**, using a custom VGA sync generator and PLL based clock division on a **DE0-CV FPGA**, to render **real time sprite based gameplay**.
- Built a **hardware LFSR random number generator**, using **shift register feedback logic**, to procedurally place pipe gaps and drive collision detection and scoring.

Smart Energy Monitor — **Embedded C & PCB-TH**

Embedded Energy Monitoring System - Team Project — *Oct 2025*

- Designed a dual channel energy monitoring system, using **signal conditioned ADC sampling on an ATmega328P**, to compute RMS voltage, peak current, and average power in real time.
- Built and validated a double layer PCB, using **Altium Designer** and UART based output, **to meet a ±5% full scale accuracy spec** across a 2.5 to 7.5 VA load range.

AWARDS

3rd Place in ECSE Design Competition 2025 — **Team Competition**

"Winnie the Bot" - Interactive Companion Robot — *Sep 2025*

- Contributed to an AI powered interviewer robot, using **dual ATmega328P Nano** microcontrollers **to drive face tracking, arm movement, and real time voice interaction.**
- **Prototyped and refined the robot's enclosure**, using iterative 3D printing and multiple design revisions, to deliver a compact, durable, and functional physical build.

SKILLS

Languages & Tools: C/C++, Python, VHDL, MIPS Assembly, MATLAB, Git, Embedded Linux, ROS, AWS

Hardware Tools: Altium Designer, LTSpice, Intel Quartus Prime, ModelSim, PSoC, Atmel AVR, Proteus, AutoCAD